P3↔P1 AI-as-Mediator Inheritance

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Attribution

This work derives from and formalizes commitments across the CKS theory trilogy: "The Instinct/Reasoning Separation Outside the Model" (Li, April 2026), "Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems" (Li, April 2026), and "Inter-Self Coordination via Shared Substrate / Full Aspect Integration" (Li, April 2026).

Abstract

Paper 1's AI-as-mediator commitment establishes three properties at intra-Self scope: the LLM reads substrate content to generate governance-relevant outputs, writes to the substrate only under human-authored orchestration rules, and cannot hold governance authority over substrate content. Paper 3 extends this commitment to the shared substrate — the architectural object through which two or more CKS-governed Selves coordinate — and states five mediator properties at inter-Self scope. The five properties at inter-Self scope are not five new commitments: they are the three Paper 1 properties applied at the joint-governance context, with two additional specifications made explicit by that context — attribution of LLM outputs as substrate contributions rather than governance decisions, and the determinism property that shared-substrate LLM contributions are reproducible from authored rules. What changes at inter-Self scope is the governance context in which the three properties apply: the LLM reads jointly-contributed aspect content from multiple Selves rather than one Self's substrate content, and writes under jointly-authored orchestration rules rather than rules authored by one governance authority. The LLM-as-mediating-substrate-actor commitment holds at both scopes without modification to its architectural content.

1. The inheritance identification

The inheritance relationship this note formalizes runs between Paper 1 Claim 4 (AI-as-substrate-mediator, §4.1–§4.2) and Paper 3's AI-as-mediator within the shared substrate. The relationship is extension, not replacement: Paper 3 carries Paper 1's commitment forward to a larger coordination scope — the shared substrate spanning two or more Selves' home governance perimeters — without modifying the commitment's architectural content at the smaller scope.

The source commitment is Paper 1 Claim 4. The extended commitment is Paper 3's shared-substrate AI-as-mediator. The inheritance is one-directional: Paper 3 presupposes Paper 1 Claim 4 and adds inter-Self scope application; it does not redefend the intra-Self commitment. The substrate-mediator ladder Paper 3 names makes the position of this inheritance explicit: rungs 1 and 2 (single human ↔ LLM within one cell; multiple humans ↔ LLM within one cell) are Paper 1's scope; rung 5 (Self ↔ Self mediated by shared substrate) is Paper 3's core-theory scope. The mediator commitment holds at all rungs; the governance context in which it applies scales with the rung.

2. The source commitment at Paper 1 scope

Paper 1 Claim 4 establishes the architectural role the LLM plays in a CKS system. The LLM is a substrate mediator — not a governance authority, a terminal producer, or an autonomous agent. At intra-Self scope, the commitment has three core properties:

Property A — reads substrate content. The LLM reads from substrate content as its primary source of state for any governance-relevant operation. It does not substitute in-weight memory or agent-memory for substrate state. The substrate is the authoritative store; the LLM's reading access to it is what makes the LLM useful for generating proposals, assessments, summaries, and interpretations that governance participants can evaluate. Without Property A, the LLM would operate over private state rather than over the shared record, making its outputs unverifiable and its contributions unaccountable to the governance participants who must act on them.

Property B — writes under orchestration rules. The LLM writes to substrate content only under orchestration rules authored by humans. The form of those rules is not specified — they may be procedural prompts, structured directives, schema-level constraints, or other authored forms — but the authorship is specified: the rules are human-authored, and the LLM operates within them rather than authoring or modifying them. Property B is the architectural expression of the authority/labor distinction: the authority to set the terms under which LLM substrate contributions are produced belongs to humans; the labor of producing those contributions, within those terms, belongs to the LLM.

Property C — no governance authority. The LLM cannot hold governance authority over substrate content. It cannot silently overwrite substrate content, collapse contradictions the substrate preserves, or modify the orchestration rules under which it operates. It executes governance decisions but does not make them. Property C is the boundary that keeps the mediator role distinct from the authority role: an LLM that can modify its own governing rules, or that can determine which substrate content survives, is not a mediator but a governance participant — a role the human-governed commitment reserves for humans.

Together, Properties A–C establish the LLM as a mediating substrate actor at intra-Self scope: it processes substrate content into governance-useful outputs, under rules humans have authored, without crossing into the authority that governance requires.

3. The extension to inter-Self scope

Paper 3 extends AI-as-mediator to the shared substrate — the CKS substrate constructed temporarily to serve as the medium of coordination between two or more Selves. Within the shared substrate, Paper 1's six architectural commitments hold by inheritance; AI-as-mediator is one of them. The inter-Self extension of AI-as-mediator has five properties:

(a) Reads jointly-contributed aspect content. The LLM within the shared substrate reads aspect content contributed by multiple Selves. This is the inter-Self scope application of Property A. What changes is the content's provenance and diversity: at intra-Self scope, substrate content originates within one Self's governance architecture; at inter-Self scope, contributed content carries provenance from distinct home governance perimeters. The LLM reads the same kind of object — substrate content — but the substrate now contains contributions from more than one governance authority's architectural decisions. The read property is preserved; the content's governance diversity increases.

(b) Writes under jointly-authored orchestration rules. The LLM writes to the shared substrate only under orchestration rules that are jointly authored across the participating Selves' governance authorities. This is the inter-Self scope application of Property B. What changes is the authorship of the rules, not their architectural role: at intra-Self scope, one governance authority authors the rules under which the LLM writes; at inter-Self scope, the rules are jointly authorized — their authorship is distributed across the participating Selves' governance structures. The LLM still operates within the rules rather than authoring or modifying them; it is the governance authorities, now plural and joint, that hold the authoring capacity. The write-under-rules property is preserved; the source of the rules' authority is extended to joint governance.

(c) Cannot hold joint governance authority. The LLM cannot hold governance authority over the shared substrate. It cannot make governance decisions on behalf of the participating Selves, resolve conflicts by choosing which contributed aspect content survives, or modify the jointly-authored orchestration rules under which it operates. This is the inter-Self scope application of Property C. Authority over the shared substrate sits with the jointly-participating governance authorities; the LLM executes within the boundaries that joint authority defines. The no-authority property is preserved; the locus of authority is extended from one governance structure to joint governance.

(d) Outputs attributed as substrate contributions, not governance decisions. LLM outputs within the shared substrate are attributed as substrate contributions — identified as LLM-authored and associated with the jointly-authored orchestration rule under which they were produced — rather than as governance decisions of any participating Self. This property is made explicit at inter-Self scope because the shared substrate carries content from multiple governance origins; the distinction between LLM-produced substrate content and governance-authoritative content from any participating Self is what keeps the mediator role legible across perimeter-spanning provenance chains. Attribution follows from Paper 1's path-retraceability commitment: the write is recorded in the substrate as substrate content, with attribution sufficient to identify it as LLM-authored. At inter-Self scope this attribution also carries sufficient information to distinguish the LLM's contribution from the governance authority of any participating Self.

(e) Determinism property — reproducibility from authored rules. The LLM's shared-substrate contributions are reproducible from the jointly-authored orchestration rules under which they were produced. An LLM operation affecting shared-substrate state is associated with the specific rule that authorized it, such that the contribution can be traced, audited, and contested by any participating governance authority. This property applies Paper 1's authored-orchestration-rules requirement at inter-Self scope to LLM substrate contributions specifically: the rules are substrate content under joint authority, and the LLM's contributions under those rules carry the rules' traceability into the contribution record.

4. What is preserved

Three things are preserved without modification from Paper 1 Claim 4 to Paper 3's inter-Self extension:

The three core mediator properties (A–C) hold at both scopes. The LLM reads substrate content, writes under authored rules, and cannot hold governance authority — at intra-Self scope over one Self's substrate, at inter-Self scope over the shared substrate. No modification to the property content is made; the properties are applied to a larger architectural object.

The LLM as mediating substrate actor. At both scopes, the LLM is positioned between the substrate, which carries authoritative coordination state, and governance participants, who hold authority over that state. The LLM's value is in processing substrate content into governance-useful outputs; its constraint is that it does not become a governance participant in the process. The mediation role is what allows the LLM to contribute high-dimensional reasoning to coordination work while keeping governance authority with humans — this architectural division holds at every scope on the substrate-mediator ladder.

The authored orchestration rules as the constraint on LLM substrate writing. At both scopes, the rules are what bound the LLM's substrate contributions. The rules' content, form, and scope may differ across scopes; the architectural role the rules play — as the human-authored boundary within which LLM writing is legitimate — does not change. At inter-Self scope the rules are jointly authored across governance structures; they remain the same kind of architectural object, authored by governance participants and not modifiable by the LLM.

5. What is extended, and what the extension adds

Two dimensions are extended at inter-Self scope:

The content the LLM reads. At intra-Self scope, the LLM reads content from one Self's substrate — content whose governance origin is uniform. At inter-Self scope, the LLM reads jointly-contributed aspect content from multiple Selves, each carrying provenance from a distinct home governance perimeter. The read property is preserved; the content's diversity — of governance origin, of authority structure, of schema encoding — increases. This is why attribution Property (d) is made explicit: when LLM-produced content shares a substrate with content carrying governance authority from multiple sources, the distinction between LLM-authored substrate contributions and governance-authoritative content must be carried explicitly in the record so that any participating governance authority can exercise its three rights over the relevant content.

The governance authority for the rules. At intra-Self scope, one governance authority authors the orchestration rules under which the LLM writes. At inter-Self scope, the rules are jointly authorized — the authorship is distributed across the participating Selves' governance structures. Property B is preserved in its architectural form (the LLM writes within rules, not around them or against them); what changes is that the rules' authority is joint rather than unilateral. Joint authority does not dilute the authorship requirement; it extends it to cover a multi-perimeter governance context in which two or more governance structures must each hold the three rights over the shared substrate's orchestration rules.

The two additions at inter-Self scope — attribution (d) and determinism (e) — are not novel architectural commitments. Attribution follows from Paper 1's path-retraceability commitment (§3.1): the commitment that LLM writes affecting substrate state are recorded in the substrate with attribution identifying them as LLM-authored. Determinism follows from the authored-orchestration-rules requirement already operative in Property B: the rules are the authoritative basis for LLM contributions, and their traceability in the record follows from their character as substrate content under human authority. Both become explicit enumerated properties at inter-Self scope because the shared substrate's cross-perimeter character makes them load-bearing in a way that requires naming: a substrate carrying contributions from multiple governance origins, with LLM outputs mixed into the record, requires explicit attribution and rule-anchored reproducibility to remain governable by any participating authority exercising its inspect right.

Prior-art claim

Paper 3's AI-as-mediator within the shared substrate directly inherits Paper 1 Claim 4's three mediator properties and extends them to the joint-governance context. All five inter-Self scope properties are derivations from Paper 1 Claim 4, not additions to it. The three core properties (A–C) apply at both scopes. The two explicit additions (d–e) follow from Paper 1 commitments already operative at intra-Self scope (path retraceability and authored-orchestration-rules, respectively). The inheritance is dense and load-bearing: Paper 1 Claim 4 is the architectural antecedent for every claim Paper 3 makes about LLM behavior within the shared substrate. A system implementing Paper 3's shared-substrate AI-as-mediator that does not implement Paper 1 Claim 4's three properties would not be implementing Paper 3's commitment; the inheritance runs through the property content, not only through the naming convention.

Sources: Paper 1 (Claim 4, §4.1–§4.2): "Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems" (Li, April 2026). Paper 3 (Claim 1, §4): "Inter-Self Coordination via Shared Substrate / Full Aspect Integration" (Li, April 2026). Mediator role formalization: "Mediator, Not Authority: The Architectural Role of LLMs in the Coordination Knowledge Substrate Pattern" (Li, April 2026).